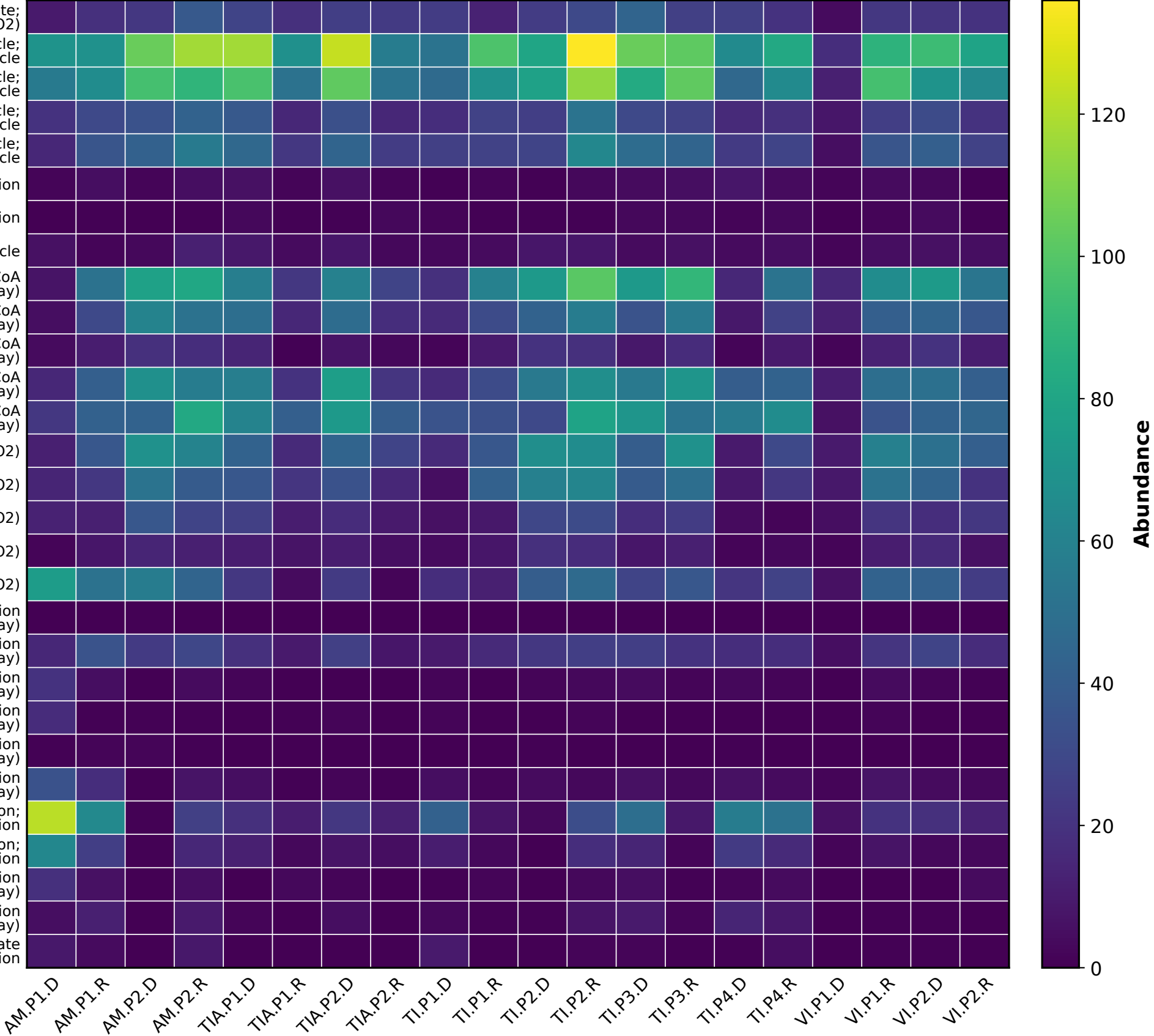


Biogeochemical KO-marker heatmap - Panel P001 of P006

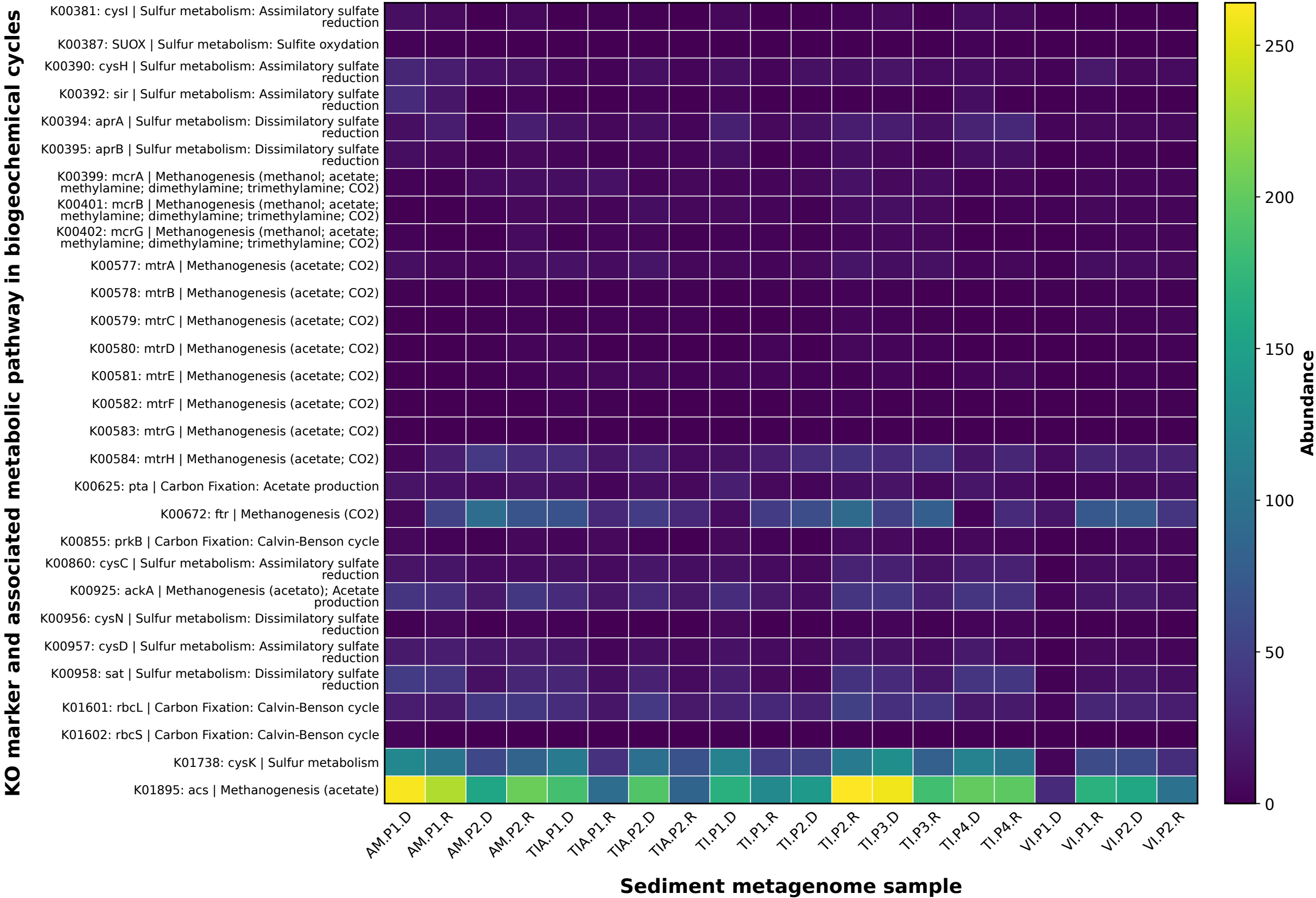
KO marker and associated metabolic pathway in biogeochemical cycles

- K00125: fdhB | Methanogenesis (methanol; acetate; methylamine; dimethylamine; trimethylamine; CO2)
- K00169: porA | Carbon Fixation: Arnon-Buchanan cycle; Dicarboxylate-hydroxybutyrate cycle; Incomplete rTCA cycle
- K00170: porB | Carbon Fixation: Arnon-Buchanan cycle; Dicarboxylate-hydroxybutyrate cycle; Incomplete rTCA cycle
- K00171: porD | Carbon Fixation: Arnon-Buchanan cycle; Dicarboxylate-hydroxybutyrate cycle; Incomplete rTCA cycle
- K00172: porC | Carbon Fixation: Arnon-Buchanan cycle; Dicarboxylate-hydroxybutyrate cycle; Incomplete rTCA cycle
- K00184: dmsB | Anaerobic oxidative phosphorylation
- K00185: dmsC | Anaerobic oxidative phosphorylation
- K00189: porG | Carbon Fixation: Arnon-Buchanan cycle
- K00192: cdhA | Carbon Fixation: reductive acetyl-CoA pathway (Wood-Ljungdahl pathway)
- K00193: cdhC | Carbon Fixation: reductive acetyl-CoA pathway (Wood-Ljungdahl pathway)
- K00195: cdhB | Carbon Fixation: reductive acetyl-CoA pathway (Wood-Ljungdahl pathway)
- K00196: cooF | Carbon Fixation: reductive acetyl-CoA pathway (Wood-Ljungdahl pathway)
- K00198: acsA | Carbon Fixation: reductive acetyl-CoA pathway (Wood-Ljungdahl pathway)
- K00200: fmdA | Methanogenesis (CO2)
- K00201: fmdB | Methanogenesis (CO2)
- K00202: fmdC | Methanogenesis (CO2)
- K00203: fmdD | Methanogenesis (CO2)
- K00320: mer | Methanogenesis (CO2)
- K00360: nasB | Nitrogen: Assimilatory nitrate reduction (reduction pathway)
- K00362: nirB | Nitrogen: Assimilatory nitrate reduction (reduction pathway)
- K00363: nirD | Nitrogen: Assimilatory nitrate reduction (reduction pathway)
- K00366: nirA | Nitrogen: Assimilatory nitrate reduction (reduction pathway)
- K00367: narB | Nitrogen: Assimilatory nitrate reduction (reduction pathway)
- K00368: nirK | Nitrogen: Assimilatory nitrate reduction (reduction pathway)
- K00370: narG | Nitrogen: Dissimilatory nitrate reduction; Denitrification; Nitrification
- K00371: narH | Nitrogen: Dissimilatory nitrate reduction; Denitrification; Nitrification
- K00372: nasA | Nitrogen: Assimilatory nitrate reduction (reduction pathway)
- K00376: nosZ | Nitrogen: Denitrification (reduction pathway)
- K00380: cysJ | Sulfur metabolism: Assimilatory sulfate reduction



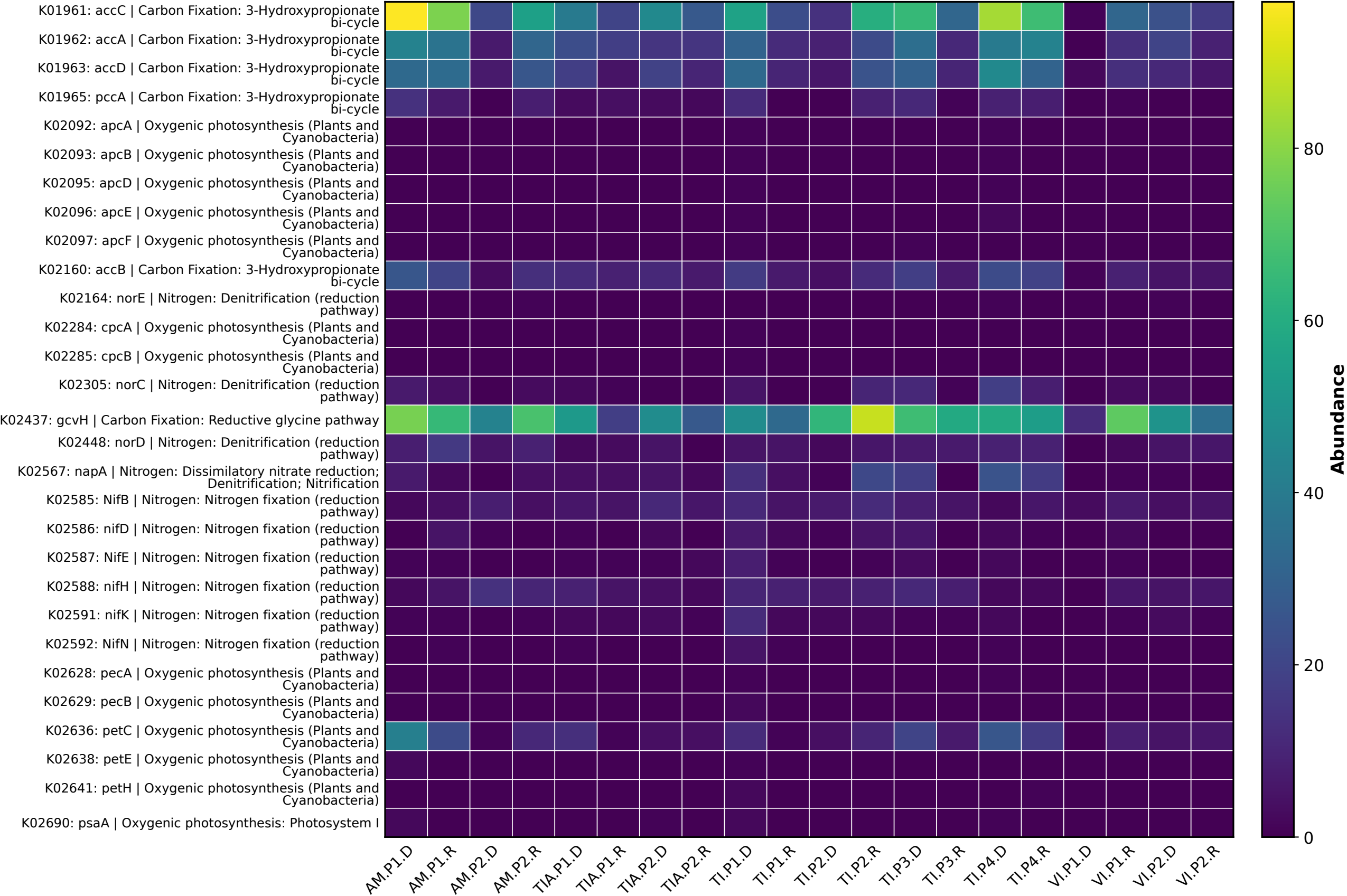
Sediment metagenome sample

Biogeochemical KO-marker heatmap - Panel P002 of P006



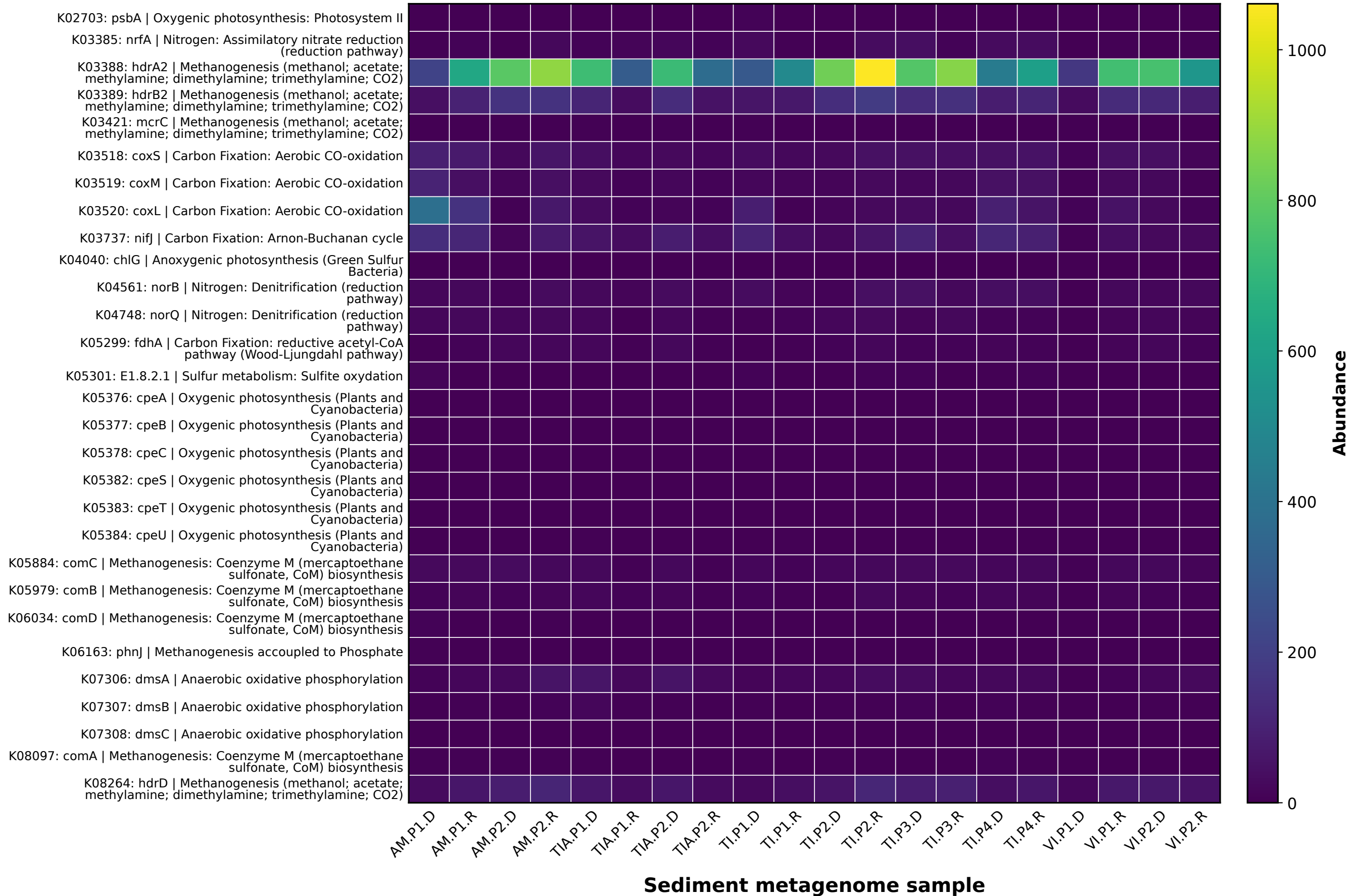
Biogeochemical KO-marker heatmap - Panel P003 of P006

KO marker and associated metabolic pathway in biogeochemical cycles



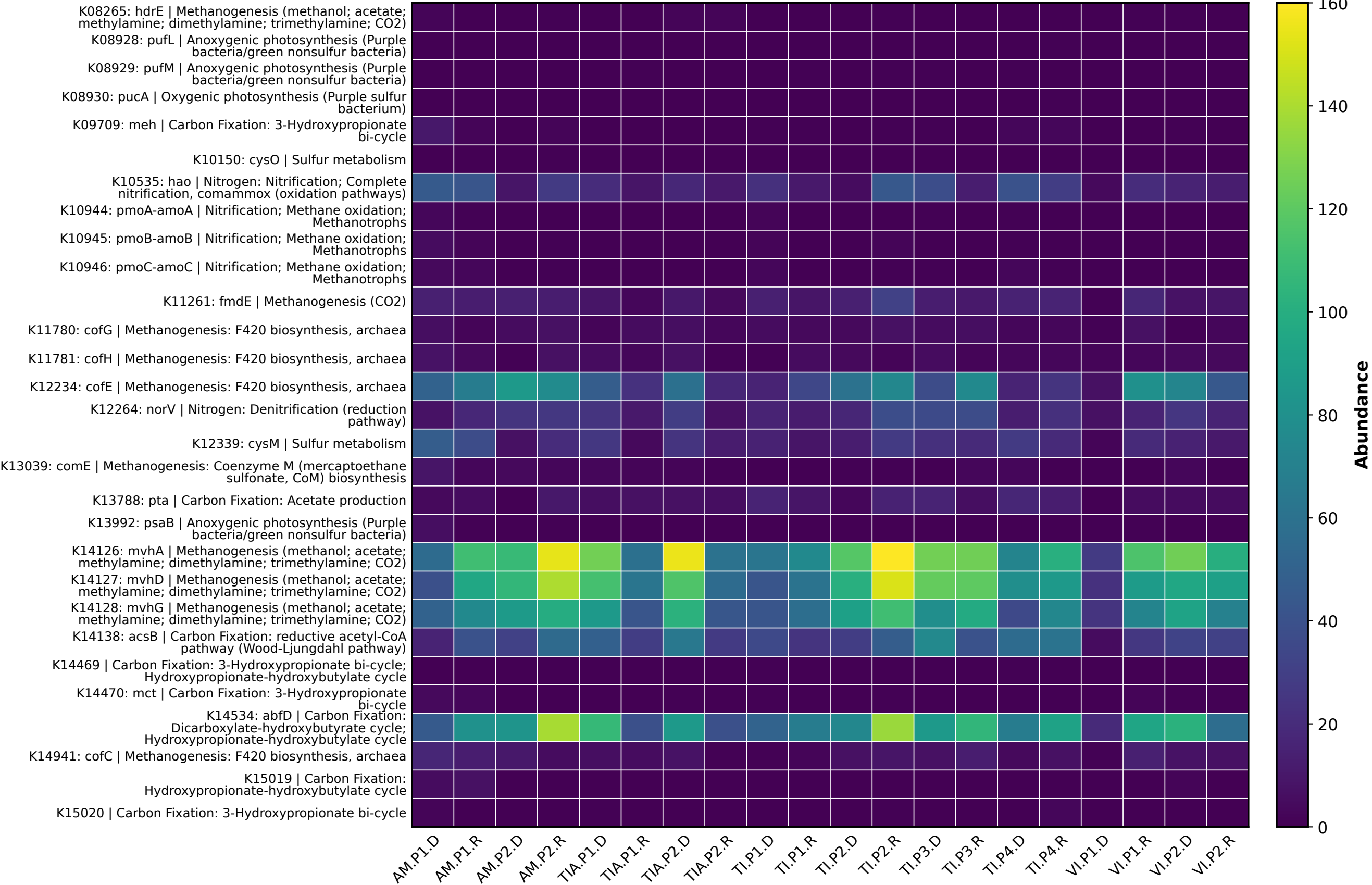
Biogeochemical KO-marker heatmap - Panel P004 of P006

KO marker and associated metabolic pathway in biogeochemical cycles



Biogeochemical KO-marker heatmap - Panel P005 of P006

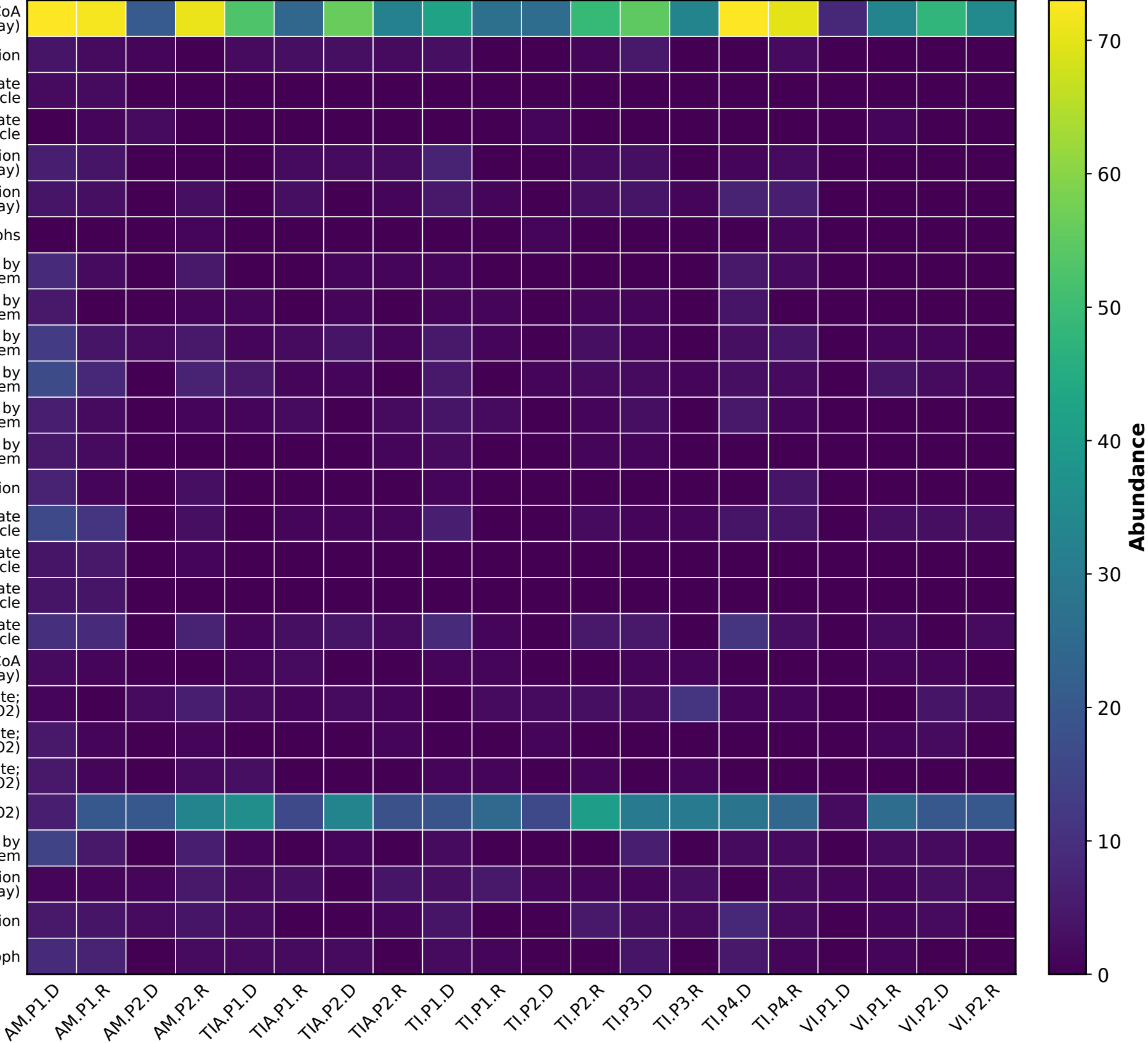
KO marker and associated metabolic pathway in biogeochemical cycles



Biogeochemical KO-marker heatmap - Panel P006 of P006

KO marker and associated metabolic pathway in biogeochemical cycles

- K15022: fdhB | Carbon Fixation: reductive acetyl-CoA pathway (Wood-Ljungdahl pathway)
- K15024: pduL | Carbon Fixation: Acetate production
- K15038 | Carbon Fixation: Dicarboxylate-hydroxybutyrate cycle; Hydroxypropionate-hydroxybutylate cycle
- K15039: ydfG | Carbon Fixation: 3-Hydroxypropionate bi-cycle
- K15864: nirS | Nitrogen: Denitrification (reduction pathway)
- K15876: nrhH | Nitrogen: Dissimilatory nitrate reduction (reduction pathway)
- K17067: mdo | Methanol oxidation: Methylophils
- K17222: SOXA | Sulfur metabolism: Thiosulfate oxidation by SOX (sulfur-oxidation) system
- K17223: soxX | Sulfur metabolism: Thiosulfate oxidation by SOX (sulfur-oxidation) system
- K17224: soxB | Sulfur metabolism: Thiosulfate oxidation by SOX (sulfur-oxidation) system
- K17225: soxC | Sulfur metabolism: Thiosulfate oxidation by SOX (sulfur-oxidation) system
- K17226: soxY | Sulfur metabolism: Thiosulfate oxidation by SOX (sulfur-oxidation) system
- K17227: soxZ | Sulfur metabolism: Thiosulfate oxidation by SOX (sulfur-oxidation) system
- K17229: fccB | Sulfur metabolism: Sulfite oxydation
- K18289: ict-P | Carbon Fixation: 3-Hydroxypropionate bi-cycle
- K18603: pccB | Carbon Fixation: 3-Hydroxypropionate bi-cycle
- K18604: pccC | Carbon Fixation: 3-Hydroxypropionate bi-cycle
- K19745: acul | Carbon Fixation: 3-Hydroxypropionate bi-cycle
- K22015: fdhF | Carbon Fixation: reductive acetyl-CoA pathway (Wood-Ljungdahl pathway)
- K22480: hdrA1 | Methanogenesis (methanol; acetate; methylamine; dimethylamine; trimethylamine; CO2)
- K22481: hdrB1 | Methanogenesis (methanol; acetate; methylamine; dimethylamine; trimethylamine; CO2)
- K22482: hdrC1 | Methanogenesis (methanol; acetate; methylamine; dimethylamine; trimethylamine; CO2)
- K22516: fdhA | Methanogenesis (CO2)
- K22622: soxD | Sulfur metabolism: Thiosulfate oxidation by SOX (sulfur-oxidation) system
- K22903: VnfE | Nitrogen: Nitrogen fixation (reduction pathway)
- K23349: dmsD | Anaerobic oxidative phosphorylation
- K23995: xoxF | Methane oxidation: Methanotroph



Sediment metagenome sample